

ANISH ANIL SHENDE

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Summary

B.Tech student in Electronics Engineering (VLSI Design and Technology) at VIT Chennai with hands-on experience across the full chip design lifecycle – RTL design and microarchitecture, functional and formal verification (UVM, SystemVerilog), and ASIC physical design (RTL-to-GDSII). Experienced in FPGA-based hardware accelerators, embedded systems, and independent end-to-end project ownership. Passionate about building efficient, verifiable, and manufacturable digital hardware.

Skills

Core Concepts	RTL Design, Microarchitecture, Digital Logic Design, ASIC Flow (RTL-to-GDSII), Physical Design, Static Timing Analysis (STA), Design Verification, Low-Power Design
HDLs & Programming	Verilog, SystemVerilog, C, C++, Python, TCL, Embedded C
EDA Tools	Cadence Genus/Innovus/Virtuoso/Tempus; Synopsys DC/ICC2/PrimeTime; QFLOW, Yosys, OpenROAD, Magic, Netgen
Simulation Tools	ModelSim, QuestaSim, VCS, OpenSTA, SymbiYosys, Cocotb, Xilinx Vivado

Education

B.Tech in Electronics Engineering (VLSI Design and Technology) <i>Vellore Institute of Technology, Chennai</i>	Aug 2023 – Present CGPA: 8.14/10.0
<i>Relevant Coursework:</i> Computer Architecture, VLSI Physical Design, Digital IC Design, Design Verification, CAD for IC Design Algorithms, Semiconductor Devices	

Work Experience

Intern – Owens Corning <i>Hardware Design, Robotics & Automation</i>	Summer 2026
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- Worked on hardware design initiatives for robotics and automation, programming and integrating 8 Yaskawa industrial robotic arms.
- Interfaced with PLC-based control systems for automation sequencing and I/O configuration, reducing manual setup steps.

Intern – Maven Silicon

Physical Design – SPI using QFLOW

- Executed complete RTL-to-GDSII flow (synthesis, floorplanning, placement, CTS, routing, sign-off) using QFLOW, improving post-CTS timing (WNS/TNS) using OpenSTA.
- Completed physical sign-off via DRC and LVS verification using Magic and Netgen, delivering a clean GDSII with zero DRC/LVS violations.

Intern – Edge Synapse

SNN-DDS Integration – Neuromorphic AI Edge Accelerator on FPGA

- Designed hardware Spiking Neural Network (SNN) architectures using LIF neuron models and developed multi-layer SNN modules in Verilog integrated with DDS IP.
- Verified and deployed the design on FPGA (Zynq UltraScale+ MPSoC) using Vivado, achieving stable real-time operation.

Projects

SparseFlow – Sparsity-Aware Neural Network Hardware Accelerator

RTL Design | UVM & Formal Verification | Physical Design

- Architected and implemented in RTL a hardware accelerator exploiting weight sparsity via a 64-bit zero-detection bitmap to clock-gate MAC cells; verified via UVM and formal verification, and synthesized to a DRC-clean GDSII (Yosys/OpenROAD, Nangate 45nm, 54,436 μm^2).

AMBA AXI4 UVM Verification IP

SystemVerilog | UVM | AXI4 Protocol

- Built a parameterized AMBA AXI4 UVM Verification IP with a master agent, scoreboard, golden reference model, and functional coverage, verifying a self-checking RTL slave DUT.

3×3 Box Filter – DSP & VLSI Implementation

Digital Signal Processing | RTL Design | Pipelined & Folded Architecture

- Designed a 3-stage pipelined RTL architecture (1 output/cycle in Vivado) and a folded variant for area reduction; published an IEEE-format trade-off analysis.

Handheld Multi-Game Embedded System

Embedded C | Raspberry Pi Pico | I2C | GPIO

- Developed a handheld multi-game embedded system with graphics rendering on a 128×64 SSD1306 OLED over I2C and GPIO-based button controls, optimized for real-time gameplay.

Certifications

Digital Design and Signoff Academic Curriculum v1.0	Cadence
Digital Design and Verification Academic Curriculum v1.0	Cadence
Cadence Student Ambassador	Cadence

Awards & Achievements

Built **RoboAOI** – an AI-powered robotic inspection and defect-sorting system for PCB/semiconductor manufacturing (YOLO11 defect detection, robotic arm in Genesis simulator) – solo, in the **AMD AI DevMaster 2026 Hackathon (Physical AI)**.

Campus Engagement

VIT Finance and Management (FAM) Club	Jul 2024 – Present
• Progressed from Club Member to Operations Director to Advisory over two years, owning club operations end-to-end.	